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Subject = Matrices,
Differential Equations
And Laplace Transform

Subject Code:- BMAS 1105

Q-1 (i) $A = \begin{bmatrix} 3 & -1 & 2 \\ -6 & 2 & 4 \\ -3 & 1 & 2 \end{bmatrix}$

Row operation:- $R_2 \rightarrow R_2 + 2R_1$: $\begin{bmatrix} 3 & -1 & 2 \\ 0 & 0 & 8 \\ -3 & 1 & 2 \end{bmatrix}$

$R_3 \rightarrow R_3 + R_1$: $\begin{bmatrix} 3 & -1 & 2 \\ 0 & 0 & 8 \\ 0 & 0 & 4 \end{bmatrix}$

$R_3 \rightarrow R_3 - \frac{1}{2}R_2$: $\begin{bmatrix} 3 & -1 & 2 \\ 0 & 0 & 8 \\ 0 & 0 & 0 \end{bmatrix}$

(ii) $B = \begin{bmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{bmatrix}$

$\Rightarrow$ Determinant: $(b-c)(c-a)(a-b)(a+b+c)$

a, b, c are distinct and $a+b+c \neq 0$: Rank = 3

If $a=b=c$: Rank = 1

If exactly two equal (say $a=b \neq c$) or $a+b+c=0$ with distinct values : Rank = 2

Q-2 (i) $\begin{bmatrix} -1 & 2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 2 \end{bmatrix}$

$R_2 \rightarrow R_2 + R_1$: $\begin{bmatrix} -1 & 2 & -2 \\ 0 & 4 & -1 \\ 0 & -3 & 4 \end{bmatrix}$

$R_3 \rightarrow R_3 - R_1$: $\begin{bmatrix} -1 & 2 & -2 \\ 0 & 4 & -1 \\ 0 & -3 & 4 \end{bmatrix}$

Continue to get : $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Normal form: $[I_3]$, Rank = 3

$$(ii) \begin{bmatrix} 3 & 2 & -1 \\ 4 & 2 & 6 \\ 7 & 4 & 5 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1 \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \therefore \text{Normal form: } [I_2 \ 0], \text{ Rank} = 2$$

Ans 3)

$$\begin{aligned} \bullet 2x_1 + x_2 + 2x_3 + x_4 &= 6 \\ \bullet 6x_1 - 6x_2 + 6x_3 + 12x_4 &= 36 \rightarrow x_1 - x_2 + x_3 + 2x_4 = 6 \\ \bullet 4x_1 + 3x_2 + 3x_3 - 3x_4 &= -1 \\ \bullet 2x_1 + 2x_2 - x_3 + x_4 &= 10 \end{aligned}$$

Augmented matrix & Row Reduction:

After Relation, we get:

$$x_1 = 1, x_2 = 2, x_3 = -1, x_4 = 2$$

Ans 4)

$$x + y + z = 1, 2x + y + 4z = k, 4x + y + 10z = k^2$$

for consistency, $\begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & 4 \\ 4 & 1 & 10 \end{vmatrix} = 0$ and check augmented matrix

$$\text{Determinant} = 1(10-4) - 1(20-10) + 1(2-4) = 6 - 10 - 2 = 0 \text{ (dependent)}$$

$$\text{For sol}^n: k^2 - 3k + 2 = 0 \Rightarrow k = 1 \text{ at } k = 2$$

Ans 5

$$x + ky + 3z = 0, 4x + 3y + kz = 0, 2x + y + 2z = 0$$

for non-trivial solⁿ, determinant = 0

$$\begin{vmatrix} 1 & k & 3 \\ 4 & 3 & k \\ 2 & 1 & 2 \end{vmatrix} = 0$$

$$= 1(6-k) - k(8-2k) + 3(8-6) = 6 - k - 8k + 2k^2 = 6$$

$$= 2k^2 - 9k = 0$$

$$k = 0 \Rightarrow k = \frac{9}{2}$$

Ans 6: $2x + y + 2z = 0$, $x + y + 3z = 0$, $4x + 3y + 6z = 0$
 Determinant $= 2(6-9) - 1(6+2) + 2(3-4) = 2(-3) - 1(8) + 2(-1) = -6 - 8 - 2 = -16 \neq 0$

(i) Trivial solⁿ: $\mathbf{0} \neq \mathbf{0}$

(ii) Non-trivial solⁿ: $b=0$

Ans 7

(i) $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Characteristic eqⁿ: $|A - \lambda I| = 0$

$(2-\lambda)[(1-\lambda)^2 - 1] = 0 \Rightarrow (1-\lambda)(\lambda-1)(\lambda-3) = 0$

Eigen values: $\lambda_1 = 1$, $\lambda_2 = 3$

for $\lambda = 1$: $(1, -1, 0)$ and $(1, 0, -1)$

for $\lambda = 3$: $(1, 1, 0)$

(ii) $B = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ 1 & -1 & 0 \end{bmatrix}$

Characteristic eqⁿ: $\lambda^3 + \lambda^2 - 21\lambda - 45 = 0$

Eigen values: $\lambda = -3, -3, 5$

Eigen vectors

for $\lambda = 5$: $(1, 2, -1)$

for $\lambda = -3$: $(3, 0, 1)$ and $(-1, 3, 0)$

Ans 8: Matrix U is unitary if $U^* U = U U^* = I$

$N = \begin{bmatrix} 0 & 1+2i \\ -1+2i & 0 \end{bmatrix}$

Let $U = (I - N)(I + N)^{-1}$

Show $U^* U = I$ by verifying N is skew-Hermitian ($N^* = -N$)
 then $(I - N)(I + N)^{-1}$ is unitary.

Ques 8 $y = an^2 + bn + c$ through $(10, 8), (20, 16), (30, 18)$

$$\begin{aligned} \cdot 100a + 10b + c &= 8 \\ \cdot 400a + 20b + c &= 16 \\ \cdot 900a + 30b + c &= 18 \end{aligned}$$

$$a = -\frac{1}{50}, b = \frac{6}{5}, c = 0$$

At boundary $(n=70 \text{ for } \sin \pi)$: $y = \frac{4900}{50} + \frac{420}{5} = -98 + 84 = -14$

Maximum at $n = 30, y = 18$ typical boundary height.

Ques 9, $V(t) = at^2 + bt + c$

$$V(3) = 64, V(6) = 133, V(9) = 208$$

System!

$$\begin{aligned} \cdot 9a + 3b + c &= 64 \\ \cdot 36a + 6b + c &= 133 \\ \cdot 81a + 9b + c &= 208 \end{aligned}$$

Solving $a = 1, b = 2, c = 49$

$$V(t) = t^2 + 2t + 49$$

$$V(15) = 225 + 30 + 49 = 304 \text{ milliseconds}$$

Ques 11 $x^2 y'' + xy' - 4y = 0$

Let $x = e^t, D = \frac{d}{dt}$

$$(D(D-1) + D - 4)y = 0 \rightarrow (D^2 - 4)y = 0$$

Auxiliary: $m^2 - 4 = 0 \rightarrow m = \pm 2$

$$y = c_1 x^2 + c_2 x^{-2}$$

Ques 12 $x^2 y'' + 2xy' = \log x$

let $x = e^t$

$(D^2 + D)y = t$

CF : $C_1 + C_2 e^{-1} = C_1 + \frac{C_2}{x}$

P.I : $\frac{1}{D^2 + D} = \frac{1}{D} (1 - D + \dots) + = \frac{t^2}{2} - t$

$y = C_1 + \frac{C_2}{x} + \frac{(\log x)^2}{2} - \log x$

Ques 13 $(D^3 - 1)y = (e^x + 1)^2 = e^{2x} + 2e^x + 1$

CF : $m^3 - 1 = 0 \Rightarrow m = 1, \omega, \omega^2$

GC : $C_1 e^x + C_2 e^{\omega x} + C_3 e^{\omega^2 x} + e^{-x/2} \left(C_4 \cos \frac{\sqrt{3}}{2} x + C_5 \sin \frac{\sqrt{3}}{2} x \right)$

P.I = $\frac{1}{(e^3 - 1)} (e^{2x} + 2e^x + 1) = \frac{e^{2x}}{7} + \frac{2e^x}{3} - 1$

$y = C_1 e^x + e^{-x/2} \left(C_2 \cos \frac{\sqrt{3}}{2} x + C_3 \sin \frac{\sqrt{3}}{2} x \right) + \frac{e^{2x}}{7} + \frac{2e^x}{3} - 1$

Ques 14 $(D^4 + D^2 + 16)y = 16x^2 + 256$

CF : $m^4 + m^2 + 16 = 0 \Rightarrow (m^2 + 4)^2 - 3m^2 = 0 \Rightarrow \pm (2 \pm \sqrt{3})i$

P.I : $\frac{1}{16} \left(1 + D^2 + \frac{D^4}{16} \right)^{-1} (16x^2 + 256) = x^2 - \frac{1}{8} + (0 = x^2 + \frac{127}{8})$

Ques 15 $(D^2 - 4D + 4)y = x^2 + \cos 2x + e^x$

$(D - 2)^2 y = \dots$

CF : $y_c = (C_1 + C_2 x) e^{2x}$

P.I for x^2 : $\frac{1}{4} \left(1 - D + \frac{3D^2}{4} \right) x^2 = \frac{x^2}{4} + \frac{x}{2} + \frac{3}{8}$

$$PI \text{ for } \cos 2x: \frac{\cos 2x}{-4+16+4} = \frac{\cos 2x}{10}, \dots \frac{-4\cos 2x + 8\sin 2x + 4\cos 2x}{10}$$

$$\text{Actually } \frac{1}{-4-8+4} \cos 2x = \frac{-\cos 2x}{8} = \frac{-\sin 2x}{16}$$

$$P.I. \text{ for } e^x: \frac{e^x}{14+4} = e^x.$$

$$\underline{\text{Ques 16}} \quad xy \sin 2x \, dx - (1+y^2 + \cos^2 x) \, dy = 0$$

$$y \sin 2x \, dx - \left(1+y^2 + \frac{1+\cos 2x}{2}\right) dy = 0$$

This is exact after rearrangement. solⁿ involves integrating for factor or substitution.

$$\underline{\text{Ques 17}} \quad (ny^3 + y) \, dx + 2(n^2y^2 + n + y^4) \, dy = 0$$

$$\frac{\partial M}{\partial y} = 3ny^2 + 1 \quad \text{Vs} \quad \frac{\partial N}{\partial x} = 4ny^2 + 2.$$

Not exact Try integrating factor or rearrange.

$$\underline{\text{Ques 18}} \quad (1+xy) \, dx + (1-xy) \, dy = 0$$

Substitute $v=xy$, separate variables.